Random Forests

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GENERAL ASSEMBLY

Agenda

- Define how and why decision trees can be improved using bagging and random forests.
- Learn how to build random forest models for classification and regression.
- Explain how to extract the most important predictors in a random forest model.

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Remind me...



Pros of decision trees?



Cons of decision trees?

Why Decision Trees?

Pros:

- Simple to understand
- “White box”
- Fast to fit

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Why Decision Trees?

Pros:

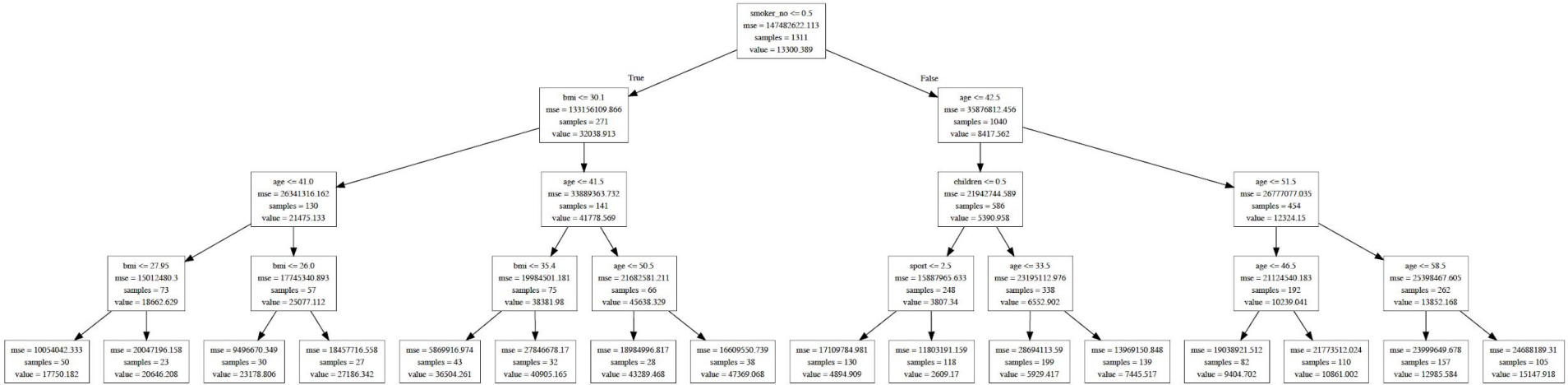
- Simple to understand
- “White box”
- Fast to fit

Cons

- High variance
- Too simple
- And about that white box…

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Something something interpretability

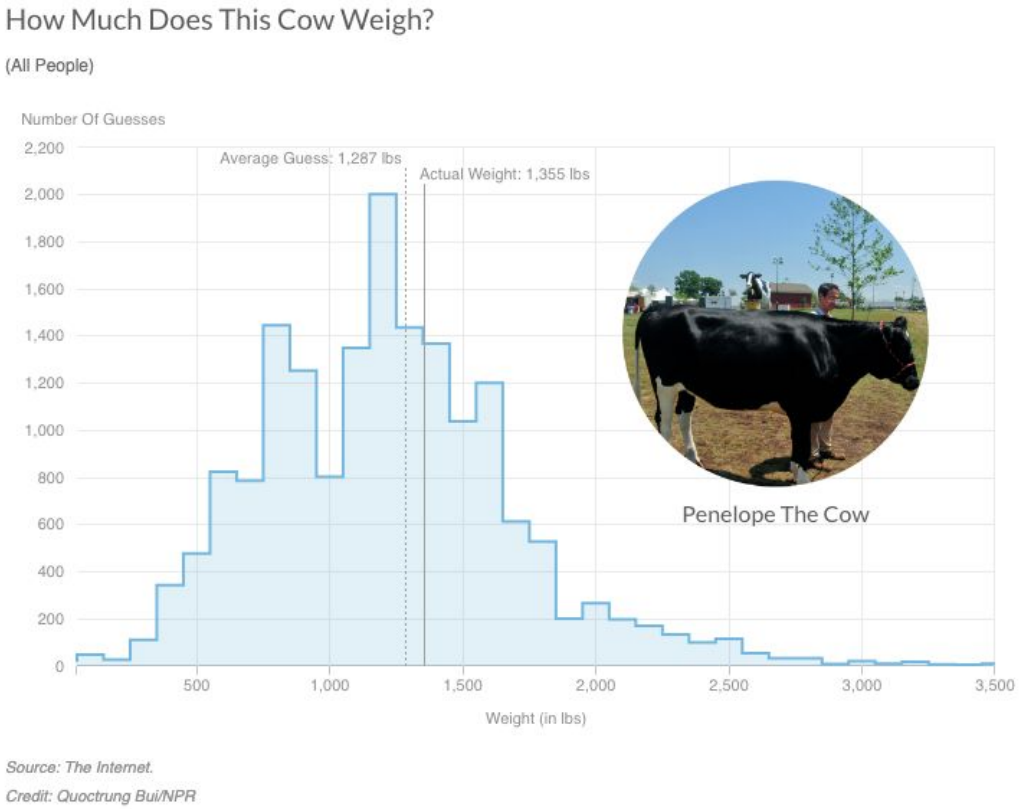


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How much does Penelope weigh?



Wisdom of the Crowds



Source: The Internet.

Credit: Quoctrung Bui/NPR

Bootstrapping

Then we introduced bagging: **b**ootstrap **agg**regat**ing**. This involved:

1. Getting a bootstrap sample
2. Fitting a model
3. Repeating 1 and 2 many times (B = 100 or 500 times)
4. The final model is the average of all B models.

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Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

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Bootstrap Simulation... Go!

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x2

x2

x2

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Bootstrap Simulation... Go!

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x3

x2

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x2

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x2

x2

Bagging

Pros:

- Ensembles trees to reduce their extremely high variance
- Averaging unbiased models results in an unbiased model… with lower variance!

Cons:

- Since the bootstrapped models are based on largely the same data, the resulting models will be *correlated!*
- Averaging correlated variables limits how much we reduce the variance by!

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The Random Forest Model

The **random forest** seeks to improve the variance reduction of bagging by reducing the tree-to-tree correlation without adding too much variance back.

We do this by adding another source of randomness:

Before each split, select a random subset of the input variables as candidates for splitting.

(Citation: ESL)

How many variables to select at each split?

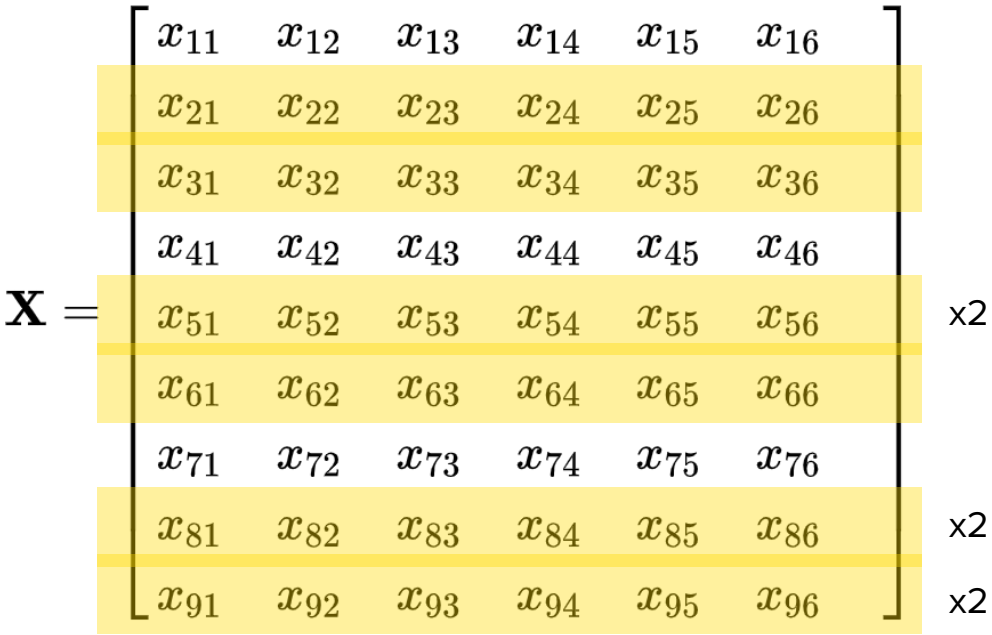
The original authors of the random forest (Breiman et al, 2001) recommend:

- $\sqrt{p}$ for classification with minimum node size of 1
- $p/3$ for regression with minimum node size of 5

And these recommendations are generally good. The authors of ESL, on the other hand, **recommend you treat them as tuning parameters** as the best value will depend on the problem.

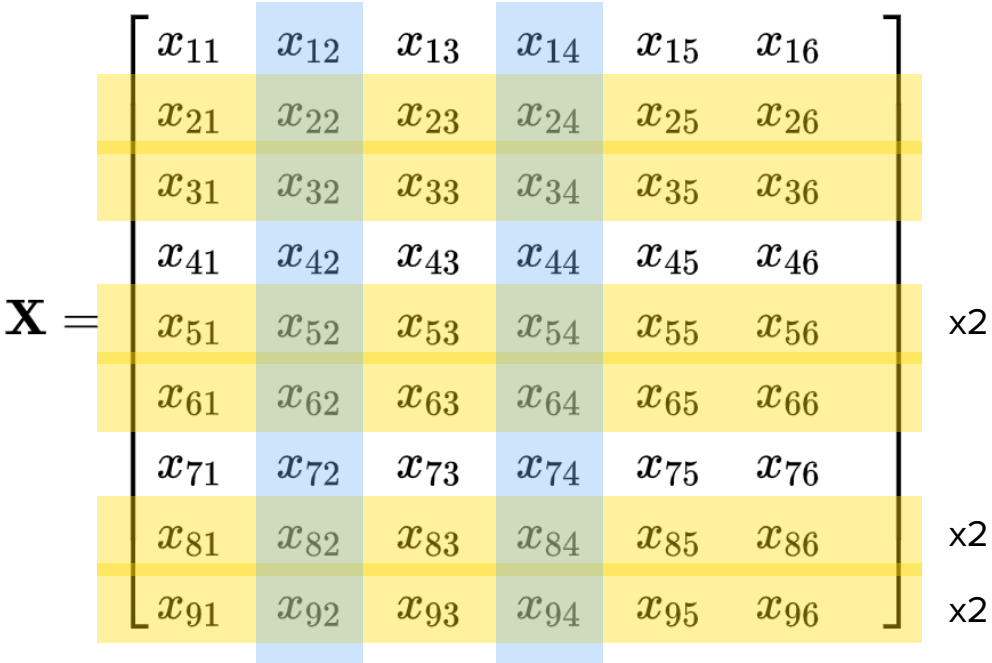
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Random Forest Simulation... Go!



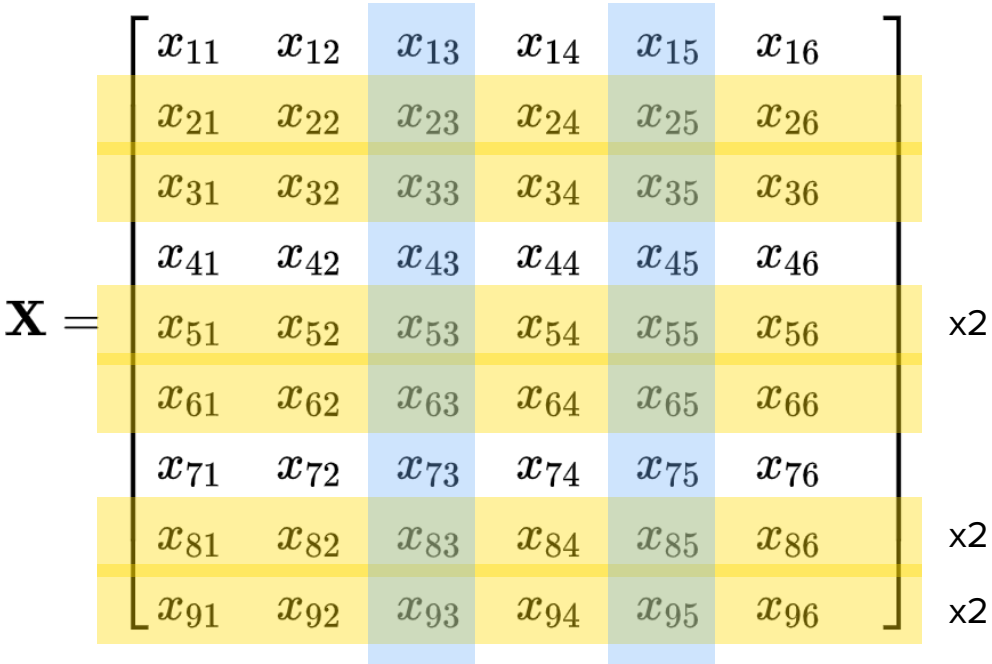
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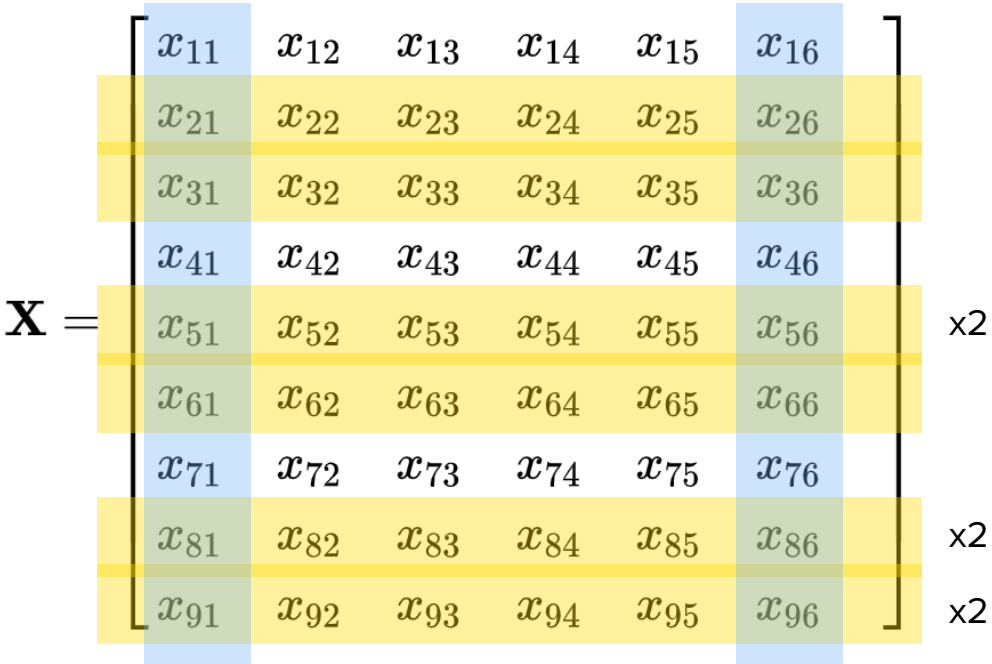
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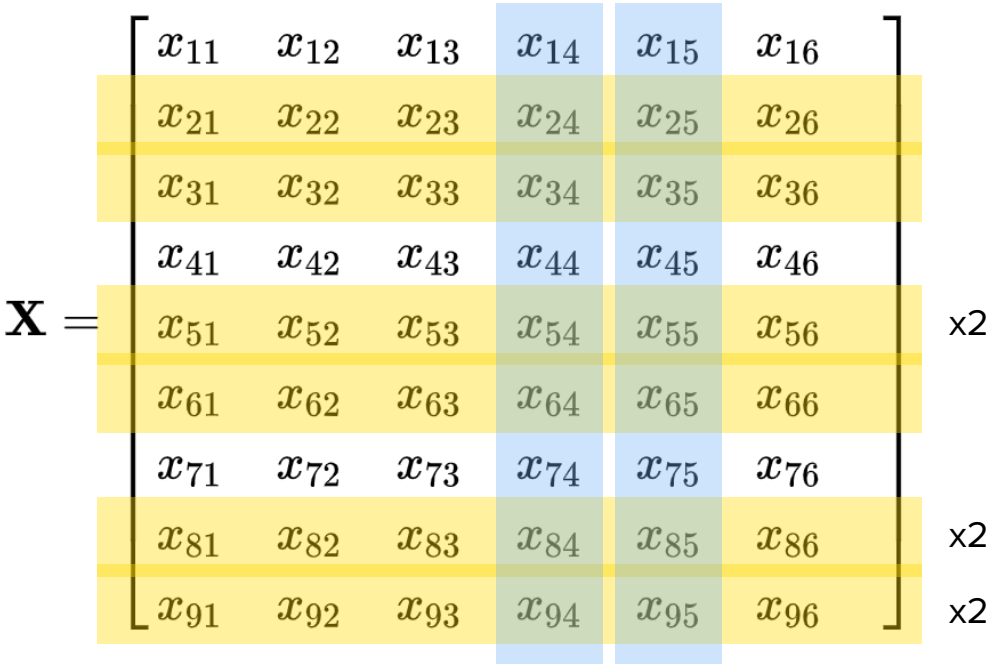
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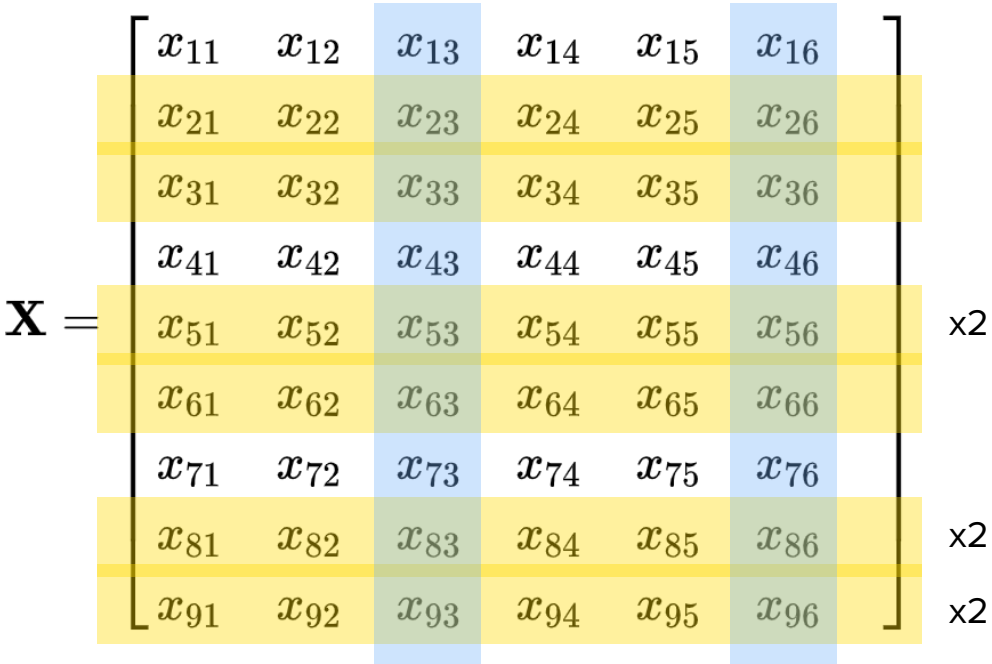
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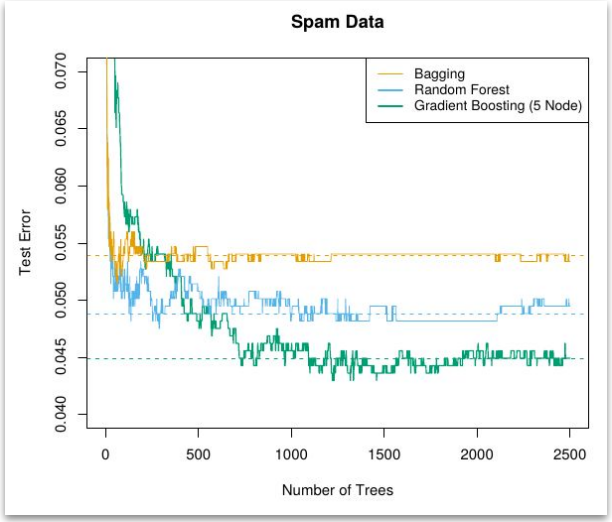
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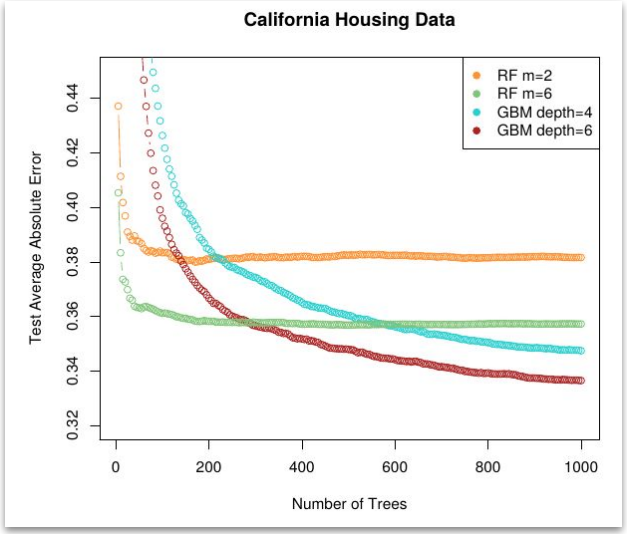
Sidenote: How many models?

In our class, I brushed this question off by saying “it should be enough so that taking the average is meaningful”. This is still true! But I want to show you actual numbers and make you reconsider our time-saving default of 100.

ESL Experiments

The authors of ESL did some experimenting to see how insensitive our results are to the number of models. Check ‘em:





Extremely Randomized Trees (Extra Trees)

There is another...

Another algorithm, called **Extremely Randomized Trees (Extra Trees)** was developed to even further reduce variance and tree-to-tree correlation at the cost of adding *a lot* more bias.

It works as follows:

Extra Trees

For some number *B* models...

1. Create a decision tree using the full dataset. (No bootstrapping!)
2. At each split, instead of computing the "best" split, **create a split totally at random! Yes, really!**
3. Repeat steps 1-2 *B* times and take the average.

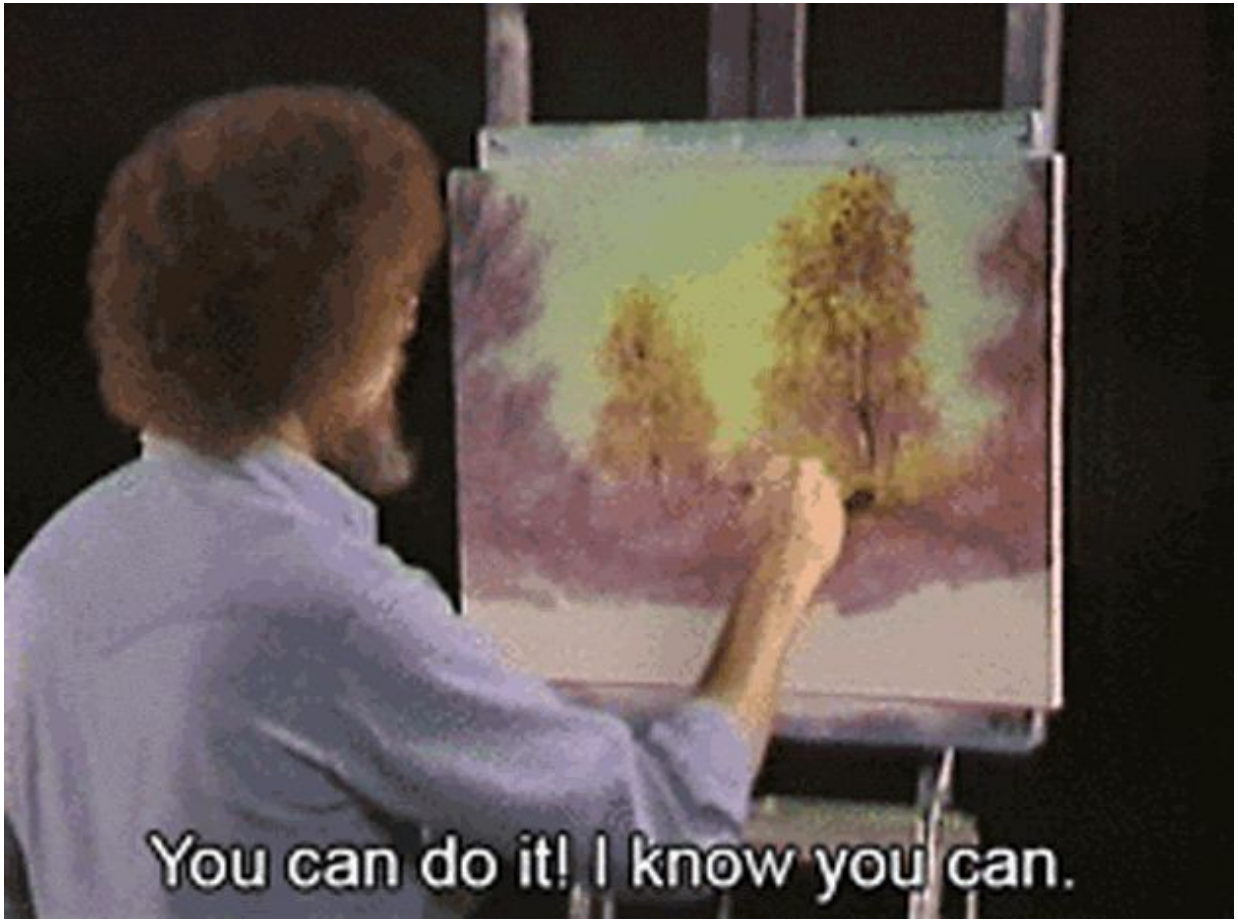
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Extra Trees

Sounds crazy, but this model actually works to some degree. The original paper (Geurts, 2005) shows a higher bias but *lower variance* than a random forest conducted on the same data.

This is likely because extra trees forces irrelevant features to be included, even if they don't help.

Further, since the optimal split is not calculated, extra trees is *much* faster than the random forest.



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Let's try them out!



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